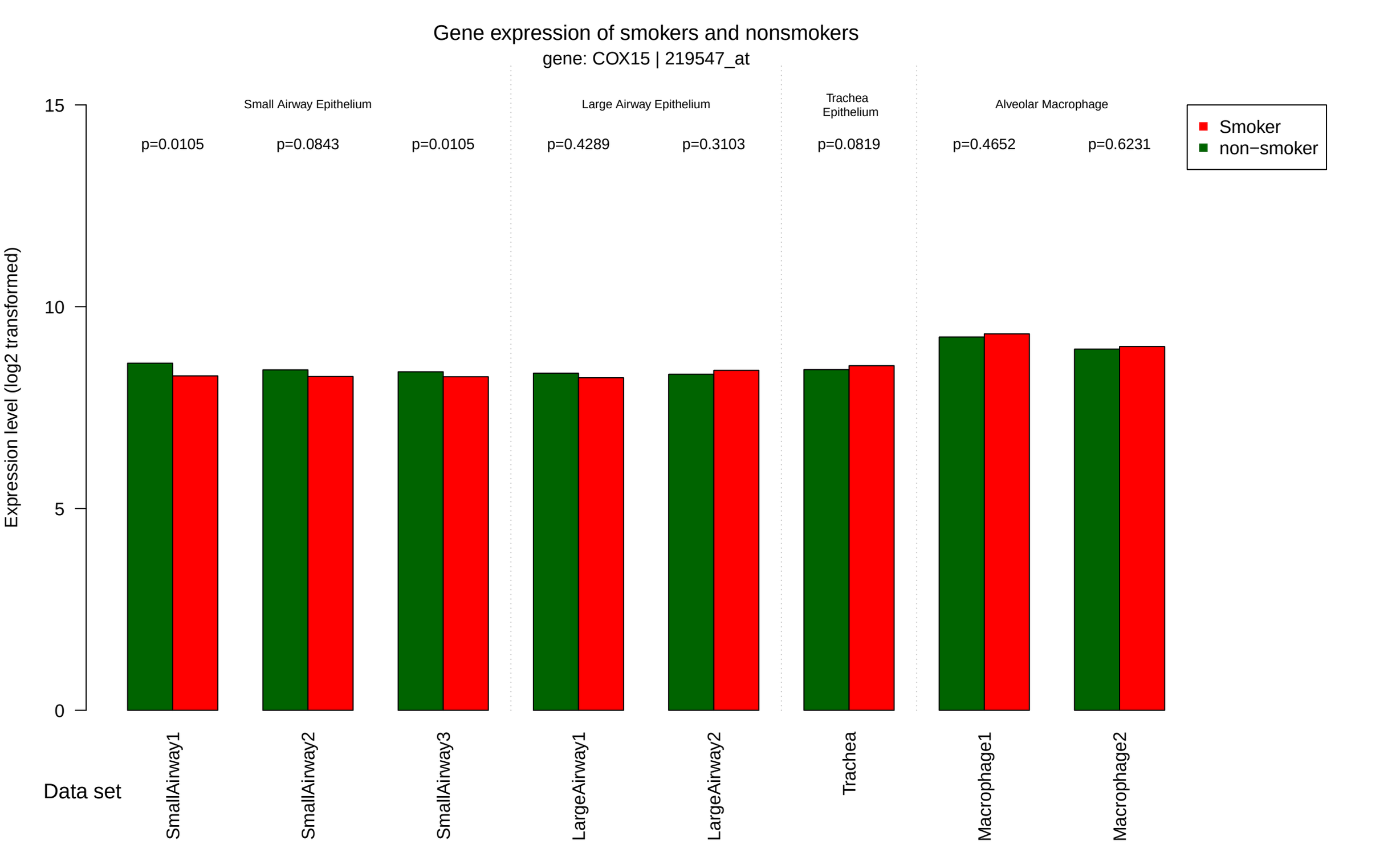
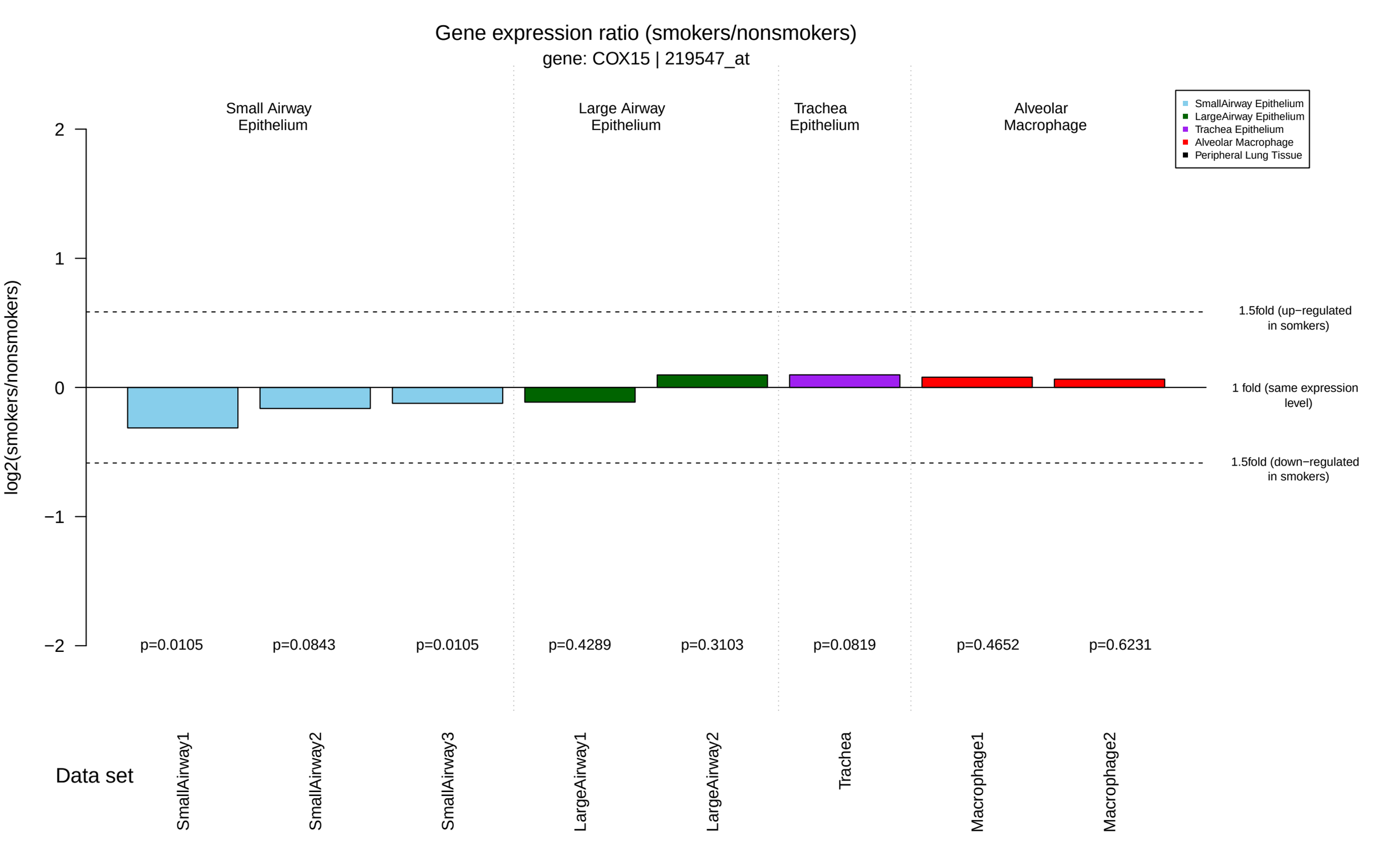
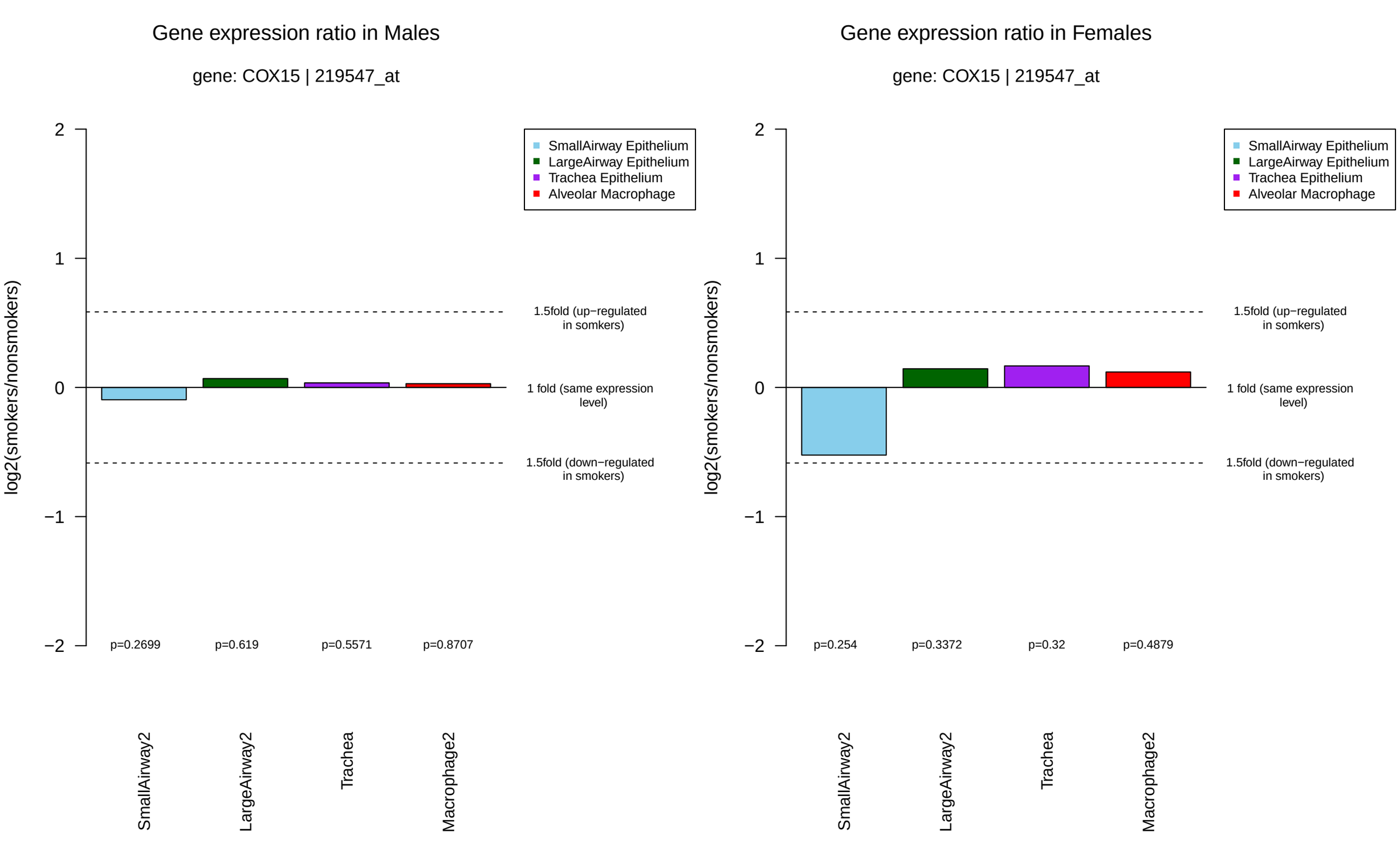
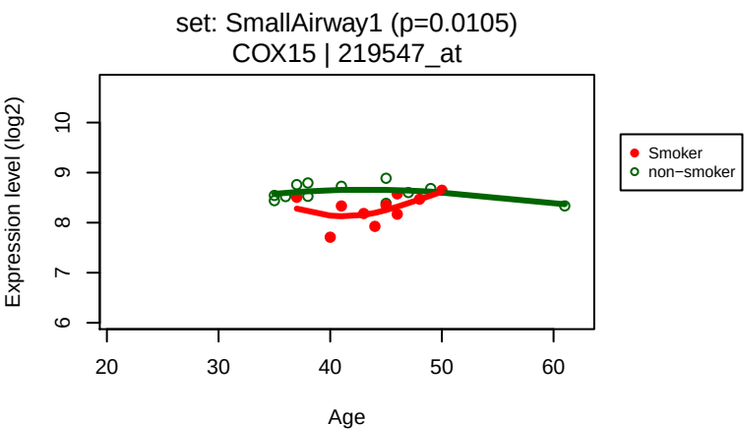
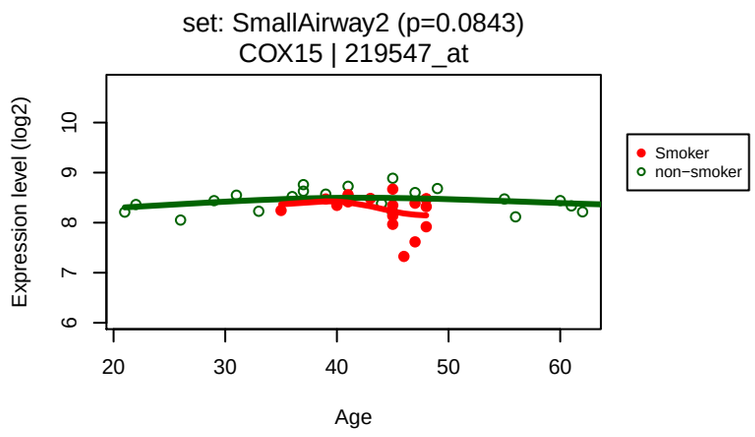
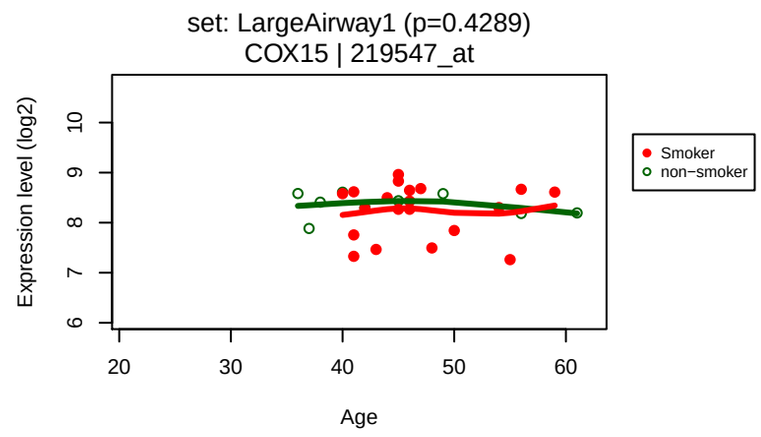

Gene expression of smokers and nonsmokers
gene: COX15 | 219547_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0105
p=0.0843
p=0.0105
p=0.4289
p=0.3103
p=0.0819
p=0.4652
p=0.6231
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

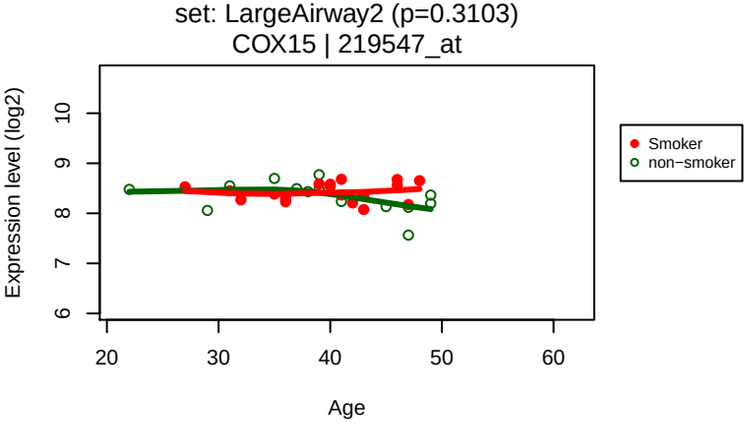

Gene expression ratio (smokers/nonsmokers)
gene: COX15 | 219547_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0105
p=0.0843
p=0.0105
p=0.4289
p=0.3103
p=0.0819
p=0.4652
p=0.6231
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

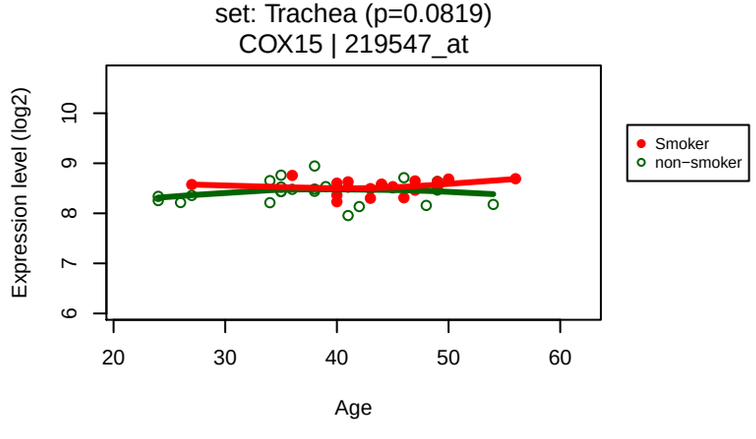

Gene expression ratio in Males
gene: COX15 | 219547_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2699
p=0.619
p=0.5571
p=0.8707
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COX15 | 219547_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.254
p=0.3372
p=0.32
p=0.4879
SmallAirway2
LargeAirway2
Trachea
Macrophage2

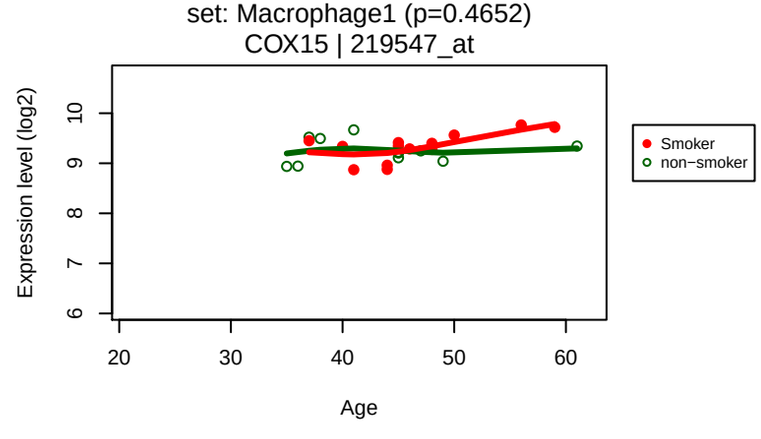

set: SmallAirway1 (p=0.0105)
COX15 | 219547_at
Expression level (log2)
Age
Smoker
non−smoker

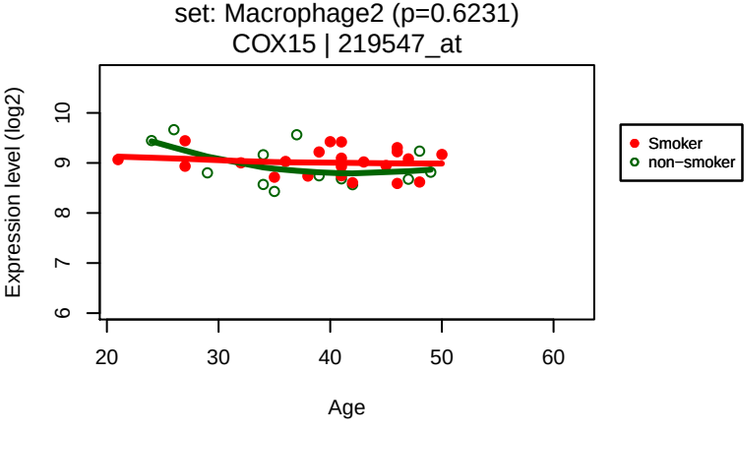

set: SmallAirway2 (p=0.0843)
COX15 | 219547_at
Expression level (log2)
Age
Smoker
non−smoker

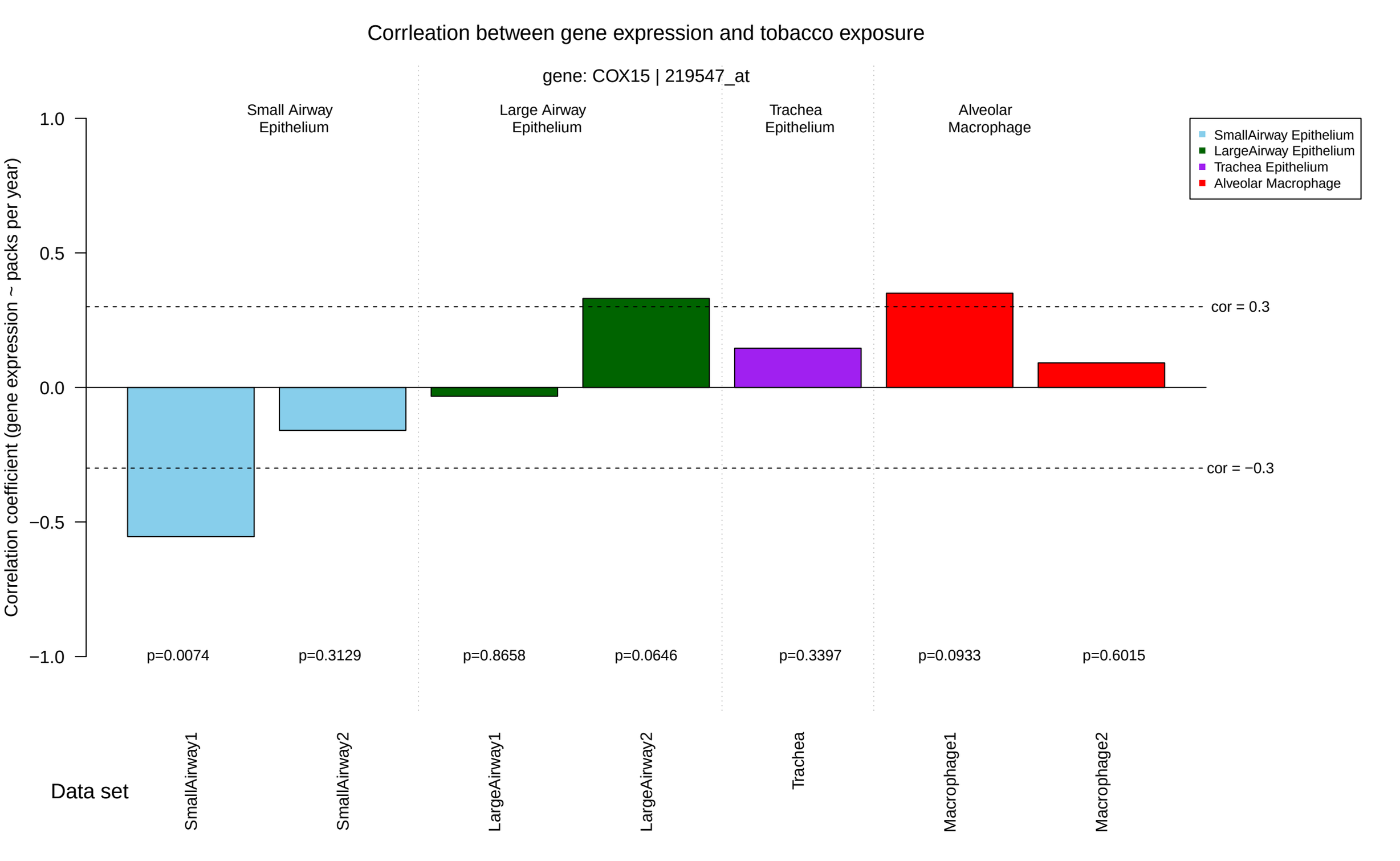

set: LargeAirway1 (p=0.4289)
COX15 | 219547_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3103)
COX15 | 219547_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0819)
COX15 | 219547_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4652)
COX15 | 219547_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6231)
COX15 | 219547_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COX15 | 219547_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0074
p=0.3129
p=0.8658
p=0.0646
p=0.3397
p=0.0933
p=0.6015
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2